

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: SIH26105

Problem Statement Title: AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform

Theme: Blockchain & Cybersecurity

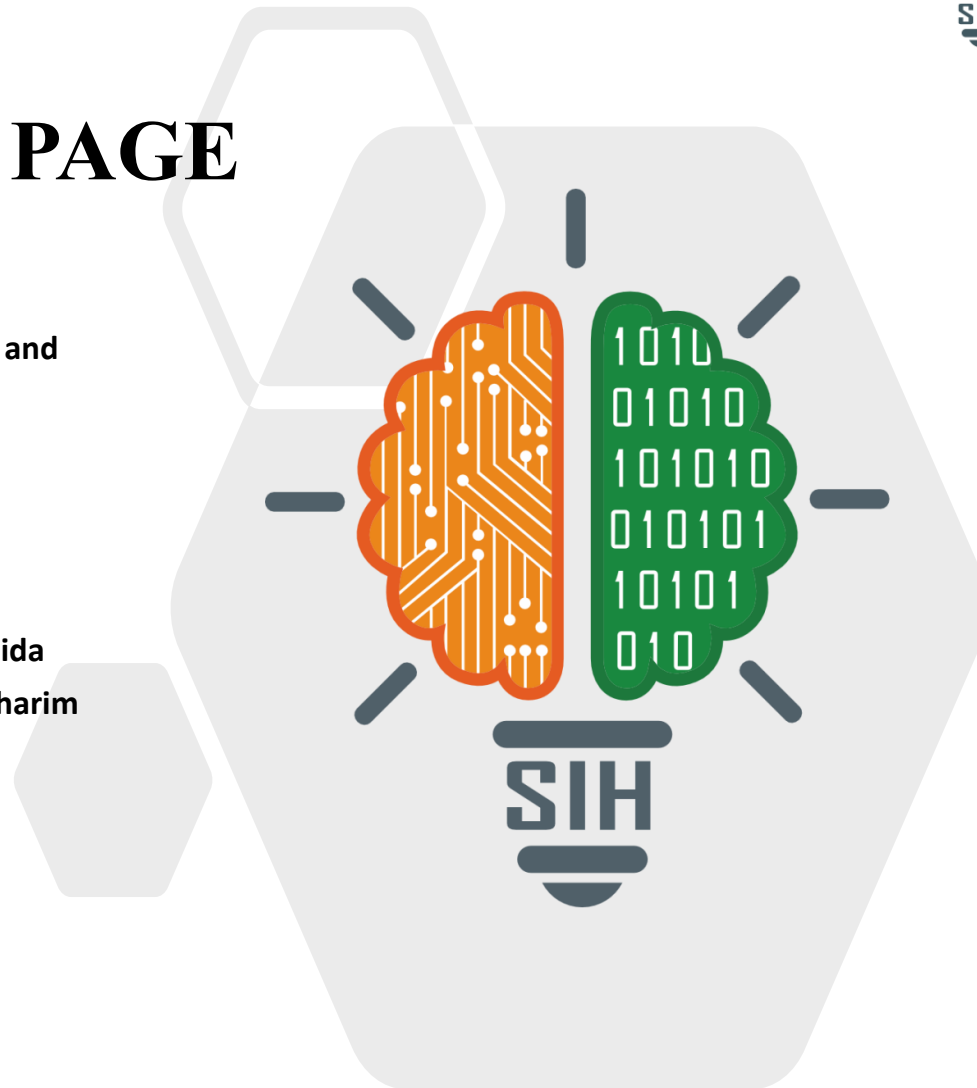
PS Category: Software

Team ID: 180219

Team Name: ByteForce_1

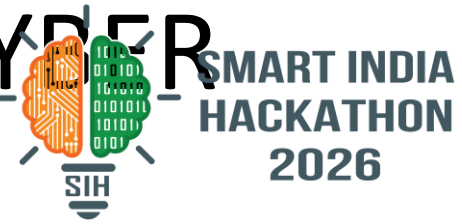
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Team Members: Aryan Baghel (Leader), Kirti, Hitesh Chauhan, Avinav Jha, Sharim Khan, Aman Sekh



IDEA: AURA SENTINEL — CONTINUOUS CYBER RISK PLATFORM

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(ID: 180219)



❖ PROPOSED SOLUTION & INNOVATION (USP)

- The Problem: Traditional scanners produce abstract 'High/Med/Low' labels. Boards and CFOs cannot understand financial liability, leading to blind security spending and stale annual compliance registers.
- Proposed Solution: AURA Sentinel ingests real-time technical security telemetry (listening sockets, process states, compute stress) and continuously calculates actuarial financial exposure in ₹ INR (Expected Annual Loss - EAL) using the Open FAIR model.
- Budget Optimization: Integrates a Pareto Knapsack optimizer to recommend highest-ROSI defense portfolios under explicit institutional budget limits (e.g. ₹1 Crore).
- Blockchain Immutability: Anchors risk evaluations, budget plans, and quarantine logs in a SHA-256 Merkle blockchain ledger, ensuring non-repudiation and satisfying CERT-In 180-day mandates.
- Core Innovation & USP: 1) Direct Actuarial Rupee Loss (EAL) quantification; 2) Mathematical Pareto ROSI optimizer; 3) Tamper-evident blockchain audit trail; 4) Zero-Trust physical presence auto-lock (YOLOv8) & touchless gestures.

TECHNICAL APPROACH & SYSTEM ARCHITECTURE



- **TECHNOLOGIES & IMPLEMENTATION METHODOLOGY**

- 3-Tier Architecture: 1) Presentation Layer: React 19 + Vite SOC Dashboard with 11 operational modules;
2) Computational Core: FastAPI Python Async Micro-Kernel on 0.0.0.0:8000;
3) Trust & AI Layer: SHA-256 Merkle Blockchain Ledger + YOLOv8 Presence Engine + SQLite 3 WAL Hub.
- Technologies Used: Python 3.10+, FastAPI, React 19, SQLite 3 (WAL Mode), Ultralytics YOLOv8-nano, Google MediaPipe 0.10, psutil, reportlab.
- Implementation Process Flow: 1. Host Ingestion: Collects active listening sockets, CPU/RAM utilization, and process states (psutil).
2. Actuarial Modeling: Computes hourly operational disruption costs and Expected Annual Loss (EAL).
3. What-If Simulation: Interactive parameter sliders and MITRE ATT&CK wargame scenarios.
4. Budget Optimization: Knapsack dynamic programming selecting optimal defense packages for maximum ROSI.
5. Blockchain Sealing: Mines SHA-256 blocks with Merkle roots & Proof-of-Work.

- **FEASIBILITY ANALYSIS & RISK STRATEGIES**

- **Technical Feasibility:** Lightweight local-first architecture designed to operate on standard Intel/AMD quad-core CPUs without requiring expensive cloud GPU servers or external subscriptions.
- **Economic Viability:** Zero recurring cloud SaaS fees. Ideal for budget-constrained academic, technical, and government institutions.
- **Current Readiness:** Fully functional standalone MVP prototype tested across 11 modules with live telemetry, Knapsack optimizer, and blockchain verification.
- **Potential Risk 1 (Telemetry Overhead):** Mitigation: Implemented asynchronous non-blocking psutil polling with 3s throttle (<1% host CPU overhead).
- **Potential Risk 2 (Subjective Risk Estimates):** Mitigation: Strict adherence to Open FAIR international standards; all formulas, baseline rates, and weights are transparent and configurable by the CISO.
- **Potential Risk 3 (Air-Gapped Compliance):** Mitigation: Operates completely offline within institutional campus LANs with zero external data leakage.

IMPACT, SCALABILITY & QUANTIFIABLE BENEFITS



- **TARGET AUDIENCE & VALUE DELIVERED**

- Target Beneficiaries: 1) AICTE & 10,000+ technical institutions across India;
2) Enterprise CISOs and IT Directors requiring board-level financial risk justification;
3) BFSI and regulated sectors needing continuous compliance posture visibility.
- Economic Impact: Eliminates wasteful security tool procurement by mathematically identifying the optimal spend zone on the Pareto frontier, maximizing Risk Reduction per Rupee invested.
- Operational Impact: Reduces periodic compliance audit preparation time from weeks of manual spreadsheet tracking to a single-click executive CISO PDF export.
- Regulatory Value: Provides cryptographic non-repudiation, giving certainty to CERT-In and RBI/SEBI auditors that historical logs have not been altered after a breach incident.
- Scalability: Modular agent architecture designed to ingest logs from enterprise forwarders (Syslog, Fluentbit, Wazuh) as campuses scale.

RESEARCH, STANDARDS & REGULATORY MAPPING



- **FRAMEWORK MAPPING & TECHNICAL REFERENCES**

- CERT-In Directions (April 2022): Mandatory 180-day preservation of cybersecurity incident logs satisfied via immutable SHA-256 Merkle blockchain anchoring.
- Digital Personal Data Protection (DPDP) Act 2023: Built-in sensitive scanner detects unmasked Aadhaar, PAN numbers, and leaked credentials.
- RBI & SEBI Cyber Resilience Frameworks: Supports continuous risk posture monitoring, sensitive port alerting, and board-level risk disclosures.
- The Open Group FAIR Standard (ANSI/FAIR): International standard for quantitative cyber risk economics and Expected Annual Loss (EAL) formulation.
- NIST CSF 2.0 & ISO/IEC 27001 & CIS Controls v8: Mapped across Identify, Protect, Detect, Respond, and Recover control functions.
- MITRE ATT&CK Matrix v15: Maps simulated wargame vectors (T1110 Brute Force, T1499 DDoS, T1046 Port Scan).
- Live Demo Playbook (5 Mins): Host Telemetry → Attack Surface → Rupee EAL → What-If Simulation → Knapsack Optimizer → Live Blockchain Tamper Showdown → PDF Audit Export.